

JOHANNESBURG INTL, South Africa

WMO: 683680

Lat: 26.143S

Lon: 28.235E

Elev: 1695

StdP: 82.55

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.4	2.2	-14.0	1.4	14.2	-12.0	1.6	13.4	10.4	9.6	9.4	10.2	4.3	210	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	29.1	14.8	28.1	14.9	27.0	15.1	19.4	23.9	18.7	23.3	18.1	22.9	4.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	16.0	21.2	17.2	15.1	20.3	16.8	14.8	20.0	63.4	24.2	61.0	23.4	59.0	22.9	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	DB	-2.5	31.1	2.1	1.2	-4.0	32.0	-5.3	32.7	-6.4	33.4	-8.0	34.3	(4)
(5)			WB	-4.4	20.4	1.7	1.2	-5.7	21.2	-6.7	21.9	-7.7	22.6	-8.9	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	19.9	19.9	18.8	16.1	13.5	11.0	10.5	13.4	16.5	18.1	18.7	19.6
	DBStd	4.28	2.08	1.97	2.13	2.45	2.94	2.56	2.81	3.33	3.59	3.21	2.84	2.23
	HDD10.0	64	0	0	0	1	7	18	28	9	2	0	0	0
	HDD18.3	1038	9	7	20	74	152	220	243	156	74	43	28	12
	CDD10.0	2369	306	278	272	184	114	48	43	114	198	252	263	298
	CDD18.3	301	57	51	33	7	1	0	0	2	21	36	41	52
	CDH23.3	2049	306	287	194	38	6	0	0	16	196	356	324	327
	CDH26.7	286	39	40	21	1	0	0	0	0	21	66	53	44
Wind	WSAvg	3.8	3.7	3.4	3.4	3.4	3.3	3.5	3.5	4.1	4.5	4.6	4.5	4.2
Precipitation	PrecAvg	727	122	97	94	47	16	8	3	7	23	76	110	110
	PrecMax	1090	338	263	334	130	99	60	22	46	175	199	230	220
	PrecMin	448	44	13	16	2	0	0	0	0	0	12	22	30
	PrecStd	144	62	53	57	31	24	14	6	10	31	45	48	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	29.8	28.9	26.0	24.1	21.0	21.1	25.1	28.8	30.2	30.1	29.9
		MCWB	16.4	15.4	14.7	14.0	11.8	8.2	8.0	11.0	12.1	13.4	14.8	15.4
	2%	DB	27.9	27.9	26.9	24.2	22.1	19.2	19.4	23.2	27.1	28.6	28.2	28.0
		MCWB	16.6	16.1	15.6	13.3	10.6	8.2	7.7	10.2	11.6	13.1	15.1	15.9
	5%	DB	26.2	26.3	25.3	23.1	20.8	18.1	18.1	21.9	25.8	27.0	26.9	26.7
		MCWB	17.0	16.5	15.4	13.2	10.5	7.8	7.5	9.7	11.5	13.5	15.1	16.3
	10%	DB	25.1	25.0	24.1	21.8	19.4	16.9	16.8	20.2	24.1	25.3	25.2	25.2
		MCWB	17.1	16.6	15.5	13.1	9.9	7.6	7.2	9.2	11.1	13.4	15.0	16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	19.7	19.7	17.1	14.6	11.2	11.0	13.7	15.2	17.5	18.7	20.2
		MCDB	25.0	24.4	22.4	21.2	19.2	16.4	17.6	19.8	22.9	23.6	24.3	24.9
	2%	WB	19.3	18.8	18.5	16.0	13.1	10.2	9.7	12.3	14.1	16.5	17.9	19.1
		MCDB	23.8	23.5	22.5	20.3	18.1	15.6	15.4	19.7	22.3	22.9	23.7	23.6
	5%	WB	18.5	18.2	17.6	15.2	12.2	9.5	8.9	11.4	13.3	15.7	17.1	18.3
		MCDB	23.1	23.0	21.9	19.4	17.6	15.0	15.0	18.7	21.3	22.3	22.9	23.0
	10%	WB	18.0	17.6	16.8	14.4	11.4	8.8	8.0	10.4	12.5	14.9	16.5	17.6
		MCDB	22.7	22.3	21.1	18.9	16.9	14.4	14.6	17.3	20.5	21.4	22.1	22.4
Mean Daily Temperature Range	5% DB	MDBR	9.7	9.9	10.0	10.3	11.1	11.6	12.0	12.5	12.9	12.2	11.0	10.2
		MCDBR	11.7	12.0	11.9	11.8	11.6	12.3	12.7	13.1	13.5	13.7	13.1	12.1
		MCWBR	4.7	4.7	4.6	4.9	5.1	5.8	5.9	5.8	5.4	5.3	4.9	4.7
	5% WB	MCDBR	9.9	9.8	9.7	9.8	10.3	10.5	11.4	12.2	12.8	12.3	11.3	10.3
	MCWBR	4.7	4.6	4.7	4.9	5.3	5.7	6.0	6.1	5.9	5.9	5.4	5.3	
Clear-Sky Solar Irradiance	taub	0.383	0.385	0.373	0.326	0.290	0.273	0.274	0.326	0.366	0.400	0.362	0.370	
	taud	2.324	2.320	2.334	2.440	2.519	2.542	2.525	2.370	2.271	2.212	2.356	2.354	
	Ebn at Noon	960	942	922	924	919	918	929	909	915	917	973	975	
	Edh at Noon	138	135	126	103	86	80	85	109	132	149	132	134	
All-Sky Solar Radiation	RadAvg	6.62	6.26	5.54	4.75	4.34	3.94	4.28	5.04	6.01	6.43	6.72	6.79	
	RadStd	0.70	0.56	0.51	0.42	0.23	0.19	0.11	0.23	0.28	0.45	0.60	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87
(47)	Regional (12 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page